If a task can issue only 240 actions, how large must the compact language model be? Count the executable actions, then search capacity and training exposure together

SentiLight: Korean utterance → five-value bulb command, reproduced by a compact language model on an edge-assisted path

1 What the task tells us

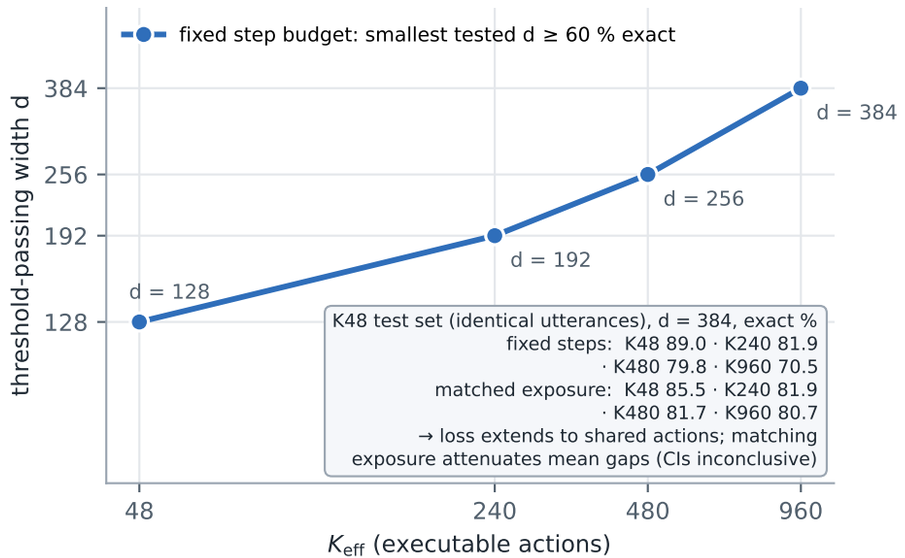
Nominal actuator space
$|H|\cdot|S|\cdot|B|\cdot|D|\cdot|CT| \approx 1.3 \times 10^{11}$ tuples

↓ executable policy π and device map f
(7 colours, 5 emotions, 5 intensities)

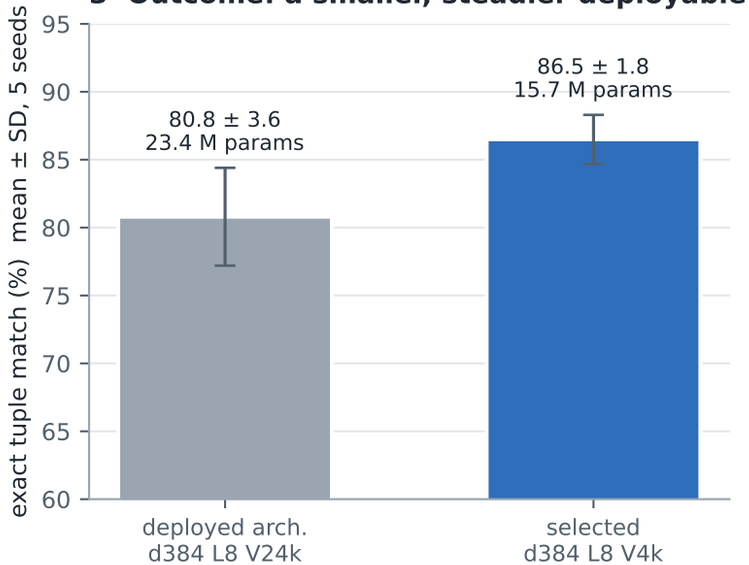
Effective action set $\mathcal{A}_{eff} = \{f(a) : a \in \mathcal{A}_{valid}(\pi)\}$
$K_{eff} = |\mathcal{A}_{eff}| = (7+1)(5+1)\times 5 = 240$, 0 collisions

K_{eff} is a task descriptor: not a hidden width, not a term in $N \approx V\cdot d + 12\cdot L\cdot d^2$. It reaches model size only through the accuracy constraint of a search over (d, L, V, budget).

2 Evidence from four nested policies (K_{eff} 48–960)



3 Outcome: a smaller, steadier deployable model



+5.8 pp (90 % CI [+1.1, +10.5]); −33 % parameters
mixed 4/6/8-bit: 17.1 MB, equivalent within ±3 pp
shared CPU (4 threads): 95 ms median
selected on mean, seed variance and downside; depth by multi-seed screening (not derivable from the policy)